

Notes: Amiti, Itskhoki, and Konings (AER, 2014)

Importers, Exporters, and Exchange Rate Disconnect

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1 Big picture

- **Question.** Why is aggregate exchange-rate pass-through into export prices incomplete, and why does pass-through vary so much across firms?
- **Core insight.** Large exporters are often large importers. When a firm's export-market exchange rate changes, its marginal cost also moves because imported inputs are priced in foreign currencies. This offsets the effect of the exchange rate on export prices.
- **Two mechanisms.** The paper identifies two channels lowering pass-through: a **marginal-cost channel** from imported inputs and a **markup channel** from variable markups and market power.
- **Key sufficient statistics.** A firm's import intensity measures how sensitive its marginal cost is to exchange rates; its export market share measures how much it can adjust markups in response to marginal-cost shocks.
- **Main empirical prediction.** Firms with high import intensity and high destination-sector market share should have low exchange-rate pass-through into destination-currency export prices.
- **Data advantage.** The authors use Belgian firm-product-destination-year export prices, firm-level imports, source countries, and destination exchange rates. This allows them to connect firm sourcing and firm pricing directly.
- **Main estimate.** Small nonimporting firms have nearly complete pass-through, while large import-intensive exporters have pass-through around 50 percent.
- **Quantitative decomposition.** The marginal-cost channel and the markup channel each explain roughly half of the difference between high-pass-through small nonimporters and low-pass-through large import-intensive exporters.
- **Aggregate implication.** The firms with the lowest pass-through account for a disproportionately large share of exports, so heterogeneity in sourcing and market shares helps explain low aggregate pass-through.
- **Takeaway.** Exchange-rate disconnect is not only about sticky prices or local-currency pricing. Even with flexible prices, imported inputs and endogenous markups can make exchange-rate pass-through incomplete.

2 Incremental contribution of the paper

2.1 Relative to standard pass-through studies

- Earlier pass-through work often estimates average pass-through using aggregate prices, product-level prices, or import/export prices without observing detailed firm sourcing.
- This paper shows that a key missing state variable is whether exporters are also importers and how much of their marginal cost is exposed to foreign exchange rates.
- It moves from average pass-through to a cross-sectional theory of pass-through based on observable firm characteristics.

Interpretation: The incremental contribution is not simply documenting incomplete pass-through. It explains why pass-through differs across exporters and why the largest exporters can have the lowest pass-through.

2.2 Relative to imported-input models

- Imported inputs had been recognized as a potential reason for incomplete pass-through into consumer prices or producer costs.

- The paper directly estimates the imported-input marginal-cost channel using firm-level import data and export price data.
- It also shows that the imported-input channel and the markup channel are not separate stories: the same firms that import intensively also tend to be large and have market power.

Interpretation: The paper provides direct micro evidence that imported inputs matter for exporter-price pass-through, not just for consumer prices or macro aggregates.

2.3 Relative to variable-markup models

- Prior work emphasizes variable markups and strategic complementarities as a source of incomplete pass-through.
- This paper adds firm-level import intensity to the variable-markup framework.
- Market share proxies for markup elasticity, while import intensity proxies for marginal-cost sensitivity.
- The two variables jointly form a sufficient statistic for firm-level pass-through in the model.

Interpretation: The novelty is the joint sufficient-statistic logic: import intensity and market share together summarize the cross-sectional determinants of pass-through within sector-destination markets.

2.4 Relative to firm heterogeneity trade models

- Existing heterogeneous-firm models show that large productive firms export more and may have different pricing behavior.
- This paper shows that large productive firms also import more intensively and source from more foreign countries.
- The model and data link firm productivity, import sourcing, market share, marginal cost, markup adjustment, and export price pass-through in one framework.

Interpretation: The paper deepens heterogeneous-firm trade models by showing that import and export decisions jointly shape pricing responses to exchange-rate shocks.

3 Data and measurement

3.1 Belgian firm-product-destination export data

- The empirical unit is a firm-product-destination-year observation.
- Export values and quantities are used to construct unit values, which proxy for export prices.
- Products are defined at HS 8-digit level; sectors are often HS 4-digit aggregates.
- The main regression sample contains over 90,000 firm-destination-product-year observations.

Interpretation: The fine unit of observation is crucial. It lets the authors compare price responses across firms selling the same or similar products to the same destinations.

3.2 Exchange rates

- The destination-specific exchange rate is denoted $e_{k,t}$.
- The empirical object is the annual log change $\Delta e_{k,t}$.

- A depreciation of the destination currency relative to the euro changes the destination-currency price of Belgian exports.
- The exchange-rate variation is common to all Belgian exporters selling to the same destination but affects firms differently depending on import intensity and market share.

Interpretation: The paper uses destination exchange-rate shocks as a common cost/demand shifter and studies heterogeneous price responses across firms.

3.3 Import intensity

- The main import-intensity variable is the share of imported intermediate inputs from outside the euro zone in total variable costs.
- Denote this variable by $\varphi_{f,t}$.
- Non-euro import intensity is used because euro-zone imports do not create euro exchange-rate exposure for Belgian firms.
- The variable is persistent across time and highly heterogeneous across exporters.

Interpretation: Import intensity is the empirical proxy for how sensitive a firm's marginal cost is to destination exchange-rate movements.

3.4 Firm market share

The firm-product-sector-destination market share is defined conceptually as

$$S_{f,s,k,t} = \frac{\text{exports of firm } f \text{ in sector } s \text{ to destination } k}{\text{total sales in sector } s \text{ to destination } k}. \quad (1)$$

- The paper constructs market shares within sector-destination-year cells.
- Market shares are meant to proxy for variable markups and markup elasticity.
- Larger market-share firms have stronger pricing-to-market incentives.

Interpretation: Market share captures the markup channel. Large market-share firms absorb marginal-cost shocks partly through markup changes rather than fully through prices.

3.5 Marginal cost proxy

The paper constructs a firm-level marginal-cost proxy from the import-weighted change in unit values of the firm's imported intermediate inputs:

$$\Delta mc_{f,t}^* \approx \sum_j \sum_m \omega_{f,j,m,t} \Delta \log U_{f,j,m,t}^* \quad (2)$$

where j indexes imported inputs, m indexes import source countries, and ω is an import expenditure share.

Interpretation: This proxy helps separate the marginal-cost channel from the markup channel. If import intensity works through marginal cost, controlling for measured marginal-cost changes should reduce the import-intensity coefficient.

4 Model and economic mechanism

4.1 Demand and market shares

The model uses a nested-CES demand structure. Each firm sells differentiated varieties within a sector-destination market. A firm's market share is denoted $S_{k,i}$ and affects its perceived demand elasticity:

$$\sigma_{k,i} = \rho(1 - S_{k,i}) + \eta S_{k,i}. \quad (3)$$

Interpretation: The firm's effective elasticity is a weighted average of within-sector and across-sector substitution elasticities. Larger firms internalize more of the sector-level price response and therefore face different markup incentives.

4.2 Markup elasticity

A firm sets a multiplicative markup over marginal cost. Let $\mathcal{M}_{k,i}$ denote the markup. Markup elasticity is

$$\Gamma_{k,i} = -\frac{\partial \log \mathcal{M}_{k,i}}{\partial \log P_{k,i}}, \quad (4)$$

holding the sector price index fixed.

- Higher market share implies a higher markup.
- Higher market share also implies a higher markup elasticity.
- Firms with high markup elasticity adjust markups more when marginal costs move.

Interpretation: This is the markup channel of incomplete pass-through. High-market-share firms absorb shocks by changing markups, reducing price pass-through.

4.3 Imported inputs and marginal cost

The firm's marginal cost depends on domestic inputs and imported inputs. A first-order approximation implies that exchange-rate movements affect marginal cost in proportion to import intensity:

$$d \log MC_i^* \approx \varphi_i d \log E_i^M + \text{other cost components}. \quad (5)$$

- φ_i is the imported input share in variable costs.
- $d \log E_i^M$ is the import-weighted exchange-rate shock faced by the firm.
- High-import-intensity exporters experience cost movements that offset the export-side exchange-rate shock.

Interpretation: If a euro depreciation raises destination-currency export prices, it may also raise the euro marginal cost of imported inputs. The net desired price adjustment is therefore smaller.

4.4 Pass-through sufficient statistic

The model implies that firm-level pass-through can be approximated by

$$\Psi_{k,i}^* = \alpha_{s,k} + \beta_{s,k} \varphi_i + \gamma_{s,k} S_{k,i}. \quad (6)$$

- $\alpha_{s,k}$ captures sector-destination average pass-through.
- $\beta_{s,k}$ captures how import intensity changes marginal-cost sensitivity.
- $\gamma_{s,k}$ captures the market-share markup channel.

Interpretation: Import intensity and market share are sufficient statistics for pass-through differences within sector-destination cells.

5 Empirical specifications

5.1 Main pass-through regression

The central empirical specification is

$$\Delta p_{f,i,k,t}^* = [\alpha_{s,k} + \beta \varphi_{f,t-1} + \tilde{\gamma} S_{f,s,k,t-1}] \Delta e_{k,t} + \text{fixed effects} + \varepsilon_{f,i,k,t}. \quad (7)$$

- $\Delta p_{f,i,k,t}^*$ is the change in producer-currency export price.
- $\Delta e_{k,t}$ is the destination exchange-rate change.
- $\Delta e_{k,t} \times \varphi_{f,t-1}$ measures how import intensity changes pass-through.
- $\Delta e_{k,t} \times S_{f,s,k,t-1}$ measures how market share changes pass-through.

Interpretation: A positive coefficient on $\Delta e \times \varphi$ in producer-currency price regressions means that import-intensive firms adjust producer prices less in destination-currency terms, i.e., have lower pass-through into destination prices.

5.2 Fixed effects

The regressions include combinations of:

- sector-destination-year fixed effects;
- sector-destination-product-year fixed effects;
- firm-product-year fixed effects;
- country-destination controls in robustness tests.

Interpretation: These fixed effects absorb common demand, cost, and product shocks. Identification comes from differential responses across firms with different import intensities and market shares facing the same exchange-rate movement.

5.3 Nonparametric bin regressions

The paper also estimates pass-through by bins of import intensity and market share:

- Import intensity is divided into quartiles.
- Firms are split into high and low market-share bins.
- The estimated pass-through coefficients are plotted and tabulated.

Interpretation: The bin approach verifies that the linear specification is not driving the result. Pass-through declines monotonically with import intensity and market share.

5.4 Mechanism tests

The paper tests the marginal-cost channel by estimating how measured marginal cost responds to exchange rates across import-intensity bins:

$$\Delta mc_{f,t}^* = a_q \Delta e_{k,t} + b_q \Delta e_{f,t}^M + u_{f,t} \quad (8)$$

for import-intensity quartiles q .

Interpretation: If import intensity proxies for marginal-cost sensitivity, the marginal cost response to exchange rates should increase with import intensity. Figure 2 shows exactly this pattern.

5.5 Which imports matter

The paper separates imports by:

- high versus low correlation between import and export exchange rates;
- high versus low pass-through into import prices;
- euro-area versus non-euro-area source countries;
- OECD versus non-OECD source countries.

Interpretation: The relevant imports are those that actually make marginal costs sensitive to the destination exchange-rate shock. The strongest effects come from imports whose exchange rates are correlated with export exchange rates and whose prices pass through strongly.

6 Identification and empirical design

6.1 Main source of identifying variation

- Exchange-rate shocks vary by destination and year.
- Import intensity varies across firms.
- Market shares vary across firm-sector-destination cells.
- The interaction between exchange-rate shocks and firm characteristics identifies heterogeneous pass-through.

Interpretation: The design is an interaction design: the same destination exchange-rate movement should affect firms differently depending on their sourcing and market power.

6.2 Why Belgian data are powerful

- Belgium is a small open economy with many importing and exporting firms.
- Firm-level customs records observe imports and exports together.
- The euro area provides a natural distinction between exchange-rate-exposed non-euro imports and less exposed euro-area imports.
- The data allow the authors to observe both sides of the firm’s international activity.

Interpretation: Most pass-through datasets observe exports but not input sourcing. This paper’s data make it possible to directly test the imported-input marginal-cost channel.

6.3 Threats and responses

- **Endogenous firm selection:** more productive firms both import and export more. The paper uses fixed effects and compares responses within narrow sector-destination cells.
- **Measurement error in prices:** unit values are imperfect price proxies. The paper uses highly disaggregated product data and robustness checks.
- **Other firm heterogeneity:** firm size, productivity, and wages may correlate with import intensity. Table 8 adds these controls.

- **Market share measure:** Belgian firm market shares may understate global market shares. Proposition 4 justifies the empirical market share measure under mild conditions.

Interpretation: The strongest evidence is the combination of direct marginal-cost tests, import-type tests, and robustness controls.

7 Tables and figures

7.1 Tables 1 and 2: exporter-importer overlap and import-intensive exporters

Read:

- Table 1 shows that 32.6 percent of manufacturing firms are exporters and/or importers; among them, 57 percent both export and import.
- Among exporters, 78.4 percent also import, while 21.6 percent only export.
- Table 2 shows that import-intensive exporters are much larger and more productive than nonimport-intensive exporters and nonexporters.
- Import-intensive exporters have 270.6 workers versus 112.2 for low-import exporters and 20.7 for nonexporters.
- They export and import from more countries and import many more HS-8 products.

Economic message: The data show that importing and exporting are tightly linked. The firms most important for aggregate exports are also firms with imported-input cost exposure.

	Exporters and/or importers	All exporters
Fraction of all firms	32.6	23.7
Among them		
exporters and importers	57.0	78.4
only exporters	15.8	21.6
only importers	27.2	—

Notes: The data include all manufacturing firms. The frequencies are averaged over the years 2000–2008.

Table 1: exporter and importer incidence

	Exporters		Non-exporters
	Import intensive	Not import intensive	
Share of imports in total cost			
All imports	0.37	0.17	0.02
Non-euro imports (φ_f)	0.17	0.01	0.00
Employment (number of workers)	270.6	112.2	20.7
Average wage rate	48.8	42.3	34.9
Material cost	103.3	28.2	3.0
Total factor productivity (log)	0.33	0.07	—
Total exports (manufacturing)	66.5	14.1	
to non-euro OECD	14.4	2.4	
Total imports (all products)	49.3	6.8	
outside euro zone	20.8	0.5	
Number of import source countries	14.4	6.6	
Number of HS-8 products imported	79.3	38.2	

Notes: The exporter subsample is split at the median of non-euro import intensity (share of non-euro imports in total costs) equal to 4.2 percent. The nonexporter subsample is all nonexporting manufacturing firms with five or more employees. Material cost, import, and export values are in millions of euros, and average wage rate is in thousands of euros. Twenty-eight percent of low import intensity firms do not import at all, and 33 percent of them do not import from outside the euro zone. The construction of the measured TFP is described in the online data Appendix.

Table 2: import-intensive exporters

7.2 Tables 3 and 4: import-intensity distribution and correlations

Read:

- Table 3 shows that 24.9 percent of exporters have zero non-euro import intensity, but the export-value distribution is skewed toward firms with higher import intensity.
- Firms with import intensity between 0.1 and 0.2 account for 23.8 percent of export value.
- Firms with import intensity between 0.2 and 0.3 account for 8.8 percent of export value.
- Table 4 shows that import intensity is positively correlated with market share, TFP, revenues, employment, and material cost.
- The highest correlation is with material cost, consistent with the interpretation that import intensity captures cost structure.

Economic message: Import intensity is highly heterogeneous among exporters and correlated with firm scale. This makes it important to distinguish the marginal-cost channel from selection and markup channels.

TABLE 3—DISTRIBUTION OF IMPORT INTENSITY AMONG EXPORTERS

	Number of firms	Fraction of firms (percent)	Fraction of export value (percent)
$\varphi_f = 0$	717	24.9	1.4
$0 < \varphi_f \leq 0.1$	1,478	51.3	38.2
$0.1 < \varphi_f \leq 0.2$	348	12.1	23.8
$0.2 < \varphi_f \leq 0.3$	155	5.4	8.8
$0.3 < \varphi_f \leq 0.4$	94	3.3	23.0
$\varphi_f > 0.4$	89	3.1	4.9

Note: Import intensity, φ_f , is the share of imported intermediate inputs from outside the euro zone in the total variable cost of the firm, averaged over the sample period.

Table 3: distribution of import intensity

	Import intensity	TFP	Revenues	Employment	Material cost
Market share	0.16	0.20	0.28	0.25	0.27
Material cost	0.23	0.70	0.99	0.83	
Employment	0.10	0.60	0.86		
Revenues	0.21	0.72			
TFP	0.15				

Notes: Cross-sectional correlations of firm-level variables averaged over time. Material costs, employment, revenues, and TFP are in logs. Import intensity is the share of imported intermediate inputs from outside the euro zone in the total cost of the firm. Market share corresponds to our measure $S_{f,k,t}$, defined in (20), aggregated to the firm-level by averaging across sector-destination-years.

Table 4: correlation structure

7.3 Table 5 and Figure 1: main pass-through result

Read:

- Table 5 reports the main pass-through regressions.
- Column 1 gives an average producer-currency pass-through coefficient of 0.203, corresponding to about 80 percent destination-currency pass-through.
- The interaction with import intensity is positive and significant: higher import intensity lowers destination-currency pass-through.
- The interaction with market share is also positive and significant: larger market-share firms have lower destination-currency pass-through.
- Figure 1 shows the nonparametric bin results: pass-through increases in producer currency, i.e. decreases in destination currency, across import-intensity quartiles.

Economic message: Import intensity and market share jointly explain large differences in pass-through. Firms with both high import intensity and high market shares have much lower pass-through.

TABLE 5—IMPORT INTENSITY, MARKET SHARE, AND PASS-THROUGH

Dependent variable: $\Delta p_{f,k,t}^*$	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\Delta e_{k,t}$	0.203*** (0.026)	0.127*** (0.027)	0.157*** (0.028)	0.149*** (0.037)	0.098*** (0.030)	0.057* (0.031)	—
$\Delta e_{k,t} \cdot \varphi_f$		0.604*** (0.112)	0.370*** (0.117)	0.341* (0.201)	0.263** (0.115)	0.473*** (0.104)	0.470** (0.236)
$\Delta e_{k,t} \cdot S_{f,k,t}$					0.238*** (0.060)	0.284*** (0.063)	0.299*** (0.100)
$\Delta mc_{f,t}^*$			0.512*** (0.030)		0.506*** (0.031)		
Fixed effects:							
$\delta_{k,t} + \delta_t$	Yes	Yes	Yes	No	Yes	Yes	No
$\delta_{k,t}$	No	No	No	No	No	No	Yes
$\delta_{f,t} + \delta_k$	No	No	No	Yes	No	No	No
Observations	93,395	93,395	93,395	93,395	93,395	93,395	93,395
R^2	0.057	0.057	0.062	0.487	0.062	0.057	0.344

Notes: Observations are at the firm-destination-product-year level. Δ corresponds to annual changes. Columns 2–3 and 5–7 include a control for the level of φ_f , and columns 5–7 also include a control for the level of the market share, $S_{f,k,t}$. Fixed effects: $\delta_{k,t} + \delta_t$ is the combination of sector-destination and year fixed effects; $\delta_{k,t}$ are sector-destination-year fixed effects; $\delta_{f,t} + \delta_k$ is the combination of firm-product-year and country fixed effects (the sector is defined at HS-4 and the product at HS-8-digit level). The results without the year fixed effects are almost identical. Standard errors are clustered at the country-year level, reported in brackets. Alternative clustering at the firm level and at the country-HS-4-digit level yield the same conclusions.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 5: import intensity, market share, and pass-through

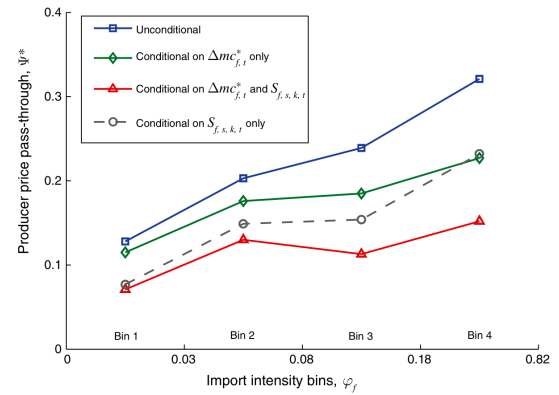


FIGURE 1. PASS-THROUGH BY QUARTILE OF IMPORT INTENSITY

Notes: Equal-sized bins in terms of firm-product-destination-year observations, sorted by φ_f . The means of φ_f in the four bins are 1.3, 5.5, 13.1, and 30.1 percent, respectively. The figure reports pass-through coefficients of $\Delta p_{f,k,t}^*$ on $\Delta e_{k,t}$ within each φ_f quartile, where the regressions include additional controls in levels and interacted with $\Delta e_{k,t}$, as indicated in the legend of the figure, to parallel columns 2–3 and 5–6 of Table 5. The pass-through coefficient in Bin 4 is significantly different from that in Bin 1 at the 1 percent level for all specifications except the third one (controlling for both $\Delta mc_{f,t}^*$ and $S_{f,k,t}$) for which it is not significant at the 5 percent level. Additional information is reported in Table A1 in the online Appendix.

Figure 1: pass-through by import-intensity quartile

7.4 Table 6 and Figure 2: bins and marginal-cost mechanism

Read:

- Table 6 splits firms by low/high import intensity and low/high market share.
- The lowest pass-through occurs in the high-import-intensity, high-market-share bin.
- This bin has a producer-currency coefficient of 0.339, implying substantially lower destination-currency pass-through.
- Figure 2 shows that marginal-cost sensitivity to exchange rates rises monotonically across import-intensity quartiles.
- The import-weighted exchange-rate response of marginal cost lies above the destination-specific exchange-rate response in every quartile.

Economic message: Table 6 confirms the joint role of sourcing and market power. Figure 2 directly supports the marginal-cost interpretation of import intensity.

	Low import intensity	High import intensity
Low market share	0.131***	0.194***
Fraction of observations	0.30	0.21
Share in export value	0.08	0.10
High market share	0.214***	0.339***
Fraction of observations	0.20	0.29
Share in export value	0.21	0.61

Notes: Coefficients from regression of $\Delta p_{f,i,k,t}^*$ on $\Delta e_{k,t}$ in respective bins without fixed effects; bin-specific intercepts included (all estimated to be close to zero). Firms are sorted by import intensity φ_f into below and above median equal to 8.1 percent, and by market share $S_{f,s,k,t}$ averaged across the years the firm served a given sector-destination into below and above the seventy-fifth percentile within each sector (HS4)-destination, which ensures a roughly even number of firm-product-destination-year observations in each high and low category. The coefficient in the high-high bin is significantly different from the other coefficients at least at a 2 percent level.

Table 6: pass-through by bins

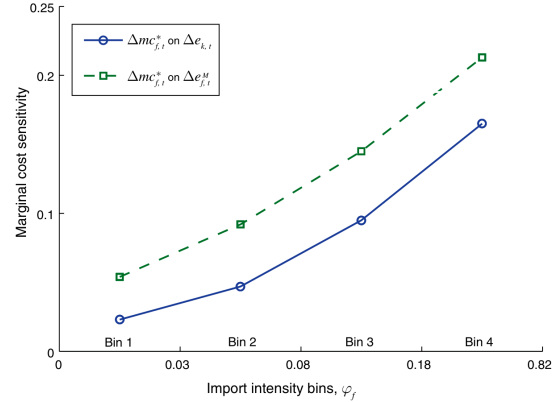


FIGURE 2. MARGINAL COST SENSITIVITY TO EXCHANGE RATES

Notes: The figure plots the pass-through coefficients from regressions of the log change in our measure of the marginal cost of the firm $\Delta mc_{f,t}^*$ on both bilateral export exchange rates $\Delta e_{k,t}$ and firm-level import-weighted exchange rate $\Delta e_{f,t}^M$, by quartiles of the φ_f distribution (as in Figure 1). Additional information is reported in Table A1 in the online Appendix.

Figure 2: marginal-cost sensitivity

7.5 Tables 7 and 8: which imports matter and robustness

Read:

- Table 7 shows that imports matter most when import exchange rates are correlated with export exchange rates and when import-price pass-through is high.
- Import intensity from within the euro area has little effect, consistent with the exchange-rate exposure interpretation.
- Non-euro imports are the relevant source of marginal-cost sensitivity.
- Table 8 shows that the main coefficients are robust to controls for firm employment, TFP, wages, and related interactions.
- The import-intensity and market-share interactions remain large and significant.

Economic message: The results are not driven by generic firm size or productivity. The relevant mechanism is the exchange-rate exposure of imported inputs.

TABLE 7—WHICH IMPORTS MATTER?

Dependent variable: $\Delta p_{f,k,t}$	Exchange rate correlation (1)	Import pass-through (2)	Euro area imports (3)	OECD versus non-OECD (4)
$\Delta e_{k,t}$	0.053* (0.031)	0.058* (0.031)	0.044 (0.040)	0.044 (0.040)
$\Delta e_{k,t} \cdot \varphi_f$	—	—	0.481*** (0.110)	—
$\Delta e_{k,t} \cdot \varphi_{f,k}^{High}$	0.864*** (0.277)	0.778*** (0.242)	—	0.472*** (0.154)
$\Delta e_{k,t} \cdot \varphi_{f,k}^{Low}$	0.376*** (0.131)	0.341 (0.241)	—	0.505*** (0.210)
$\Delta e_{k,t} \cdot \varphi_f^{Other}$	—	0.027 (0.316)	—	—
$\Delta e_{k,t} \cdot \varphi_f^{Euro}$	—	—	0.060 (0.126)	0.057 (0.126)
$\Delta e_{k,t} \cdot S_{f,k,t}$	0.284*** (0.063)	0.286*** (0.063)	0.282*** (0.063)	0.282*** (0.064)
FE: $\delta_{s,k} + \delta_t$	Yes	Yes	Yes	Yes
Observations	93,395	93,395	93,395	93,395
R^2	0.058	0.058	0.058	0.058

Notes: Results from the augmented specification

$$\Delta p_{f,k,t} = [\alpha + \beta^{High} \varphi_{f,k}^{High} + \beta^{Low} \varphi_{f,k}^{Low} + \beta^{Other} \varphi_f^{Other} + \gamma S_{f,k,t}] \Delta e_{k,t} + \dots + \epsilon_{f,k,t,t}.$$

Column 1: splits imports into high and low correlation bins for each firm-destination depending on whether the correlation (based on annual log changes) between import and export exchange rate is above 0.7; decomposition of import intensity: $\varphi_f \equiv \varphi_{f,k}^{High} + \varphi_{f,k}^{Low}$ for every destination k . Column 2: splits imports by pass-through into import prices into three bins: high (above 0.5), low (below 0.5), and other (imprecisely estimated), as explained in the text ($\varphi_f \equiv \varphi_f^{High} + \varphi_f^{Low} + \varphi_f^{Other}$). Column 3: does not split imports from outside the euro area (φ_f), but instead additionally includes import intensity from within the euro area (φ_f^{Euro}). Column 4: splits non-euro imports into imports from high-income OECD and low-income non-OECD countries ($\varphi_f \equiv \varphi_{f,k}^{High} + \varphi_{f,k}^{Low}$), also controlling for non-euro imports (φ_f^{Euro}). All regressions additionally control for the levels of all interaction terms and include country-destination and time fixed effects. Other details as in Table 5.

Table 7: which imports matter?

TABLE 8—ROBUSTNESS WITH ADDITIONAL CONTROLS

Dependent variable: $\Delta p_{f,k,t}$	(1)	(2)	(3)
$\Delta e_{k,t} \cdot \varphi_f$	0.413*** (0.106)	0.433*** (0.109)	0.418*** (0.119)
$\Delta e_{k,t} \cdot S_{f,k,t}$	0.219*** (0.065)	0.249*** (0.064)	0.245*** (0.065)
$\Delta e_{k,t} \cdot \log L_{f,t}$	0.044*** (0.012)	—	—
$\Delta e_{k,t} \cdot \log TFP_{f,t}$	—	0.070*** (0.023)	0.080*** (0.024)
$\Delta \log W_{f,t}^*$	—	—	0.004* (0.002)
$\Delta \log TFP_{f,t}$	—	—	0.035*** (0.007)
FE: $\delta_{s,k} + \delta_t$	Yes	Yes	Yes
Observations	92,576	92,106	87,608
R^2	0.058	0.058	0.061

Notes: The same specification as in column 6 of Table 5, augmented with additional controls. $L_{f,t}$ is firm employment, $W_{f,t}^*$ is firm average wage rate, and $TFP_{f,t}$ is the estimate of firm total factor productivity.

Table 8: robustness with additional controls

8 Short summary

- The paper explains incomplete exchange-rate pass-through using imported inputs and variable markups.
- Large exporters are often large importers.
- Imported inputs make marginal costs sensitive to exchange rates, offsetting export-side exchange-rate movements.
- Market share captures markup variability: high-market-share firms adjust markups more and pass through less.
- Import intensity and market share are sufficient statistics for firm-level pass-through within sector-destination cells.
- Belgian firm-product-destination data allow direct measurement of both exports and imports.
- Small nonimporting firms have almost complete pass-through.
- Large import-intensive firms have pass-through around 50 percent.
- The marginal-cost and markup channels each explain roughly half of the cross-firm pass-through difference.
- Robustness checks confirm that the effects are not driven by size, TFP, wages, or generic productivity differences.

9 Conclusion

9.1 Core conclusion

Amiti, Itskhoki, and Konings show that exchange-rate pass-through is low in part because the firms that dominate exports are also import-intensive and have market power. Their imported inputs make marginal costs move with exchange rates, while their market shares allow them to adjust markups actively. Together these mechanisms generate incomplete pass-through even with flexible pricing.

9.2 Economic intuition

When a destination currency changes, a simple exporter with domestic inputs faces a straightforward pricing problem: the exchange-rate shock changes the foreign-currency price of its goods. But an importer-exporter faces offsetting movements in marginal cost, because imported inputs become more or less expensive. If the firm also has high market share, it can absorb part of the shock in its markup. Therefore, the firms with the largest aggregate weight in exports naturally exhibit the lowest pass-through.

9.3 Takeaway

The paper’s central contribution is to connect sourcing, market power, and exchange-rate pass-through at the firm level. It shows that aggregate exchange-rate disconnect emerges from micro heterogeneity: the largest exporters are precisely the firms with the strongest imported-input and markup channels.